

Birla Institute of Technology and Science, Pilani

Quantum Machine Learning (AIMLZG545)

Assignment 2: Quantum Approach to a Classical Problem

April 22, 2026

Instructions

- Total Marks: **15**
- This is an **open-book** assignment
- This assignment builds on **Assignment 1**
- You must use the **same problem** identified earlier
- Submit:
 - PDF report (3–5 pages)
 - Code notebook (PennyLane / Qiskit)
- Recommended constraint: **Use ≤ 6 qubits**

Objective

To apply a quantum machine learning approach to a classical problem and analyze its feasibility.

1 Problem Recap (2 Marks)

- What is the problem?
- Why is it difficult classically?

2 Data Representation (3 Marks)

- What are the input features?
- How are they encoded into quantum states?
- Which encoding method is used (angle / amplitude / basis)?

3 Quantum Model Design (4 Marks)

Choose one:

- Variational Quantum Circuit (VQC)
- Quantum Kernel Method
- Draw the circuit architecture
- What gates are used?
- How is entanglement introduced?
- What is measured?

4 Implementation and Results (3 Marks)

Minimum Reporting Checklist (Compulsory):

- Circuit diagram (clearly labeled)
- Description of dataset used (size, features)
- At least one of the following plots:
 - Measurement probability distribution
 - Loss vs iterations (for VQC)
 - Kernel matrix visualization (for kernel method)
- Final output/result (e.g., prediction, classification accuracy, or kernel similarity)

Additional Expectations:

- Brief explanation of what each plot shows
- Interpretation of results (not just screenshots)

Note: The above is a **minimum requirement**. You are encouraged to include additional plots, experiments, or analysis if relevant.

5 Discussion and Limitations (3 Marks)

- Does the quantum approach provide any advantage?
- What are the limitations of your approach?
- How do NISQ constraints (noise, qubits, depth) affect your results?

6 References

Provide references used.

Evaluation Summary

- Problem Recap: 2 marks
- Data Encoding: 3 marks
- Quantum Model Design: 4 marks
- Implementation & Results: 3 marks
- Discussion: 3 marks

Note: Emphasis will be on clarity, correctness, and depth of analysis.